

Figure 1.9 Low-level specifications, written as unit tests, flow naturally from the high-level specifications.

You could also write this specification using conventional unit-testing tools, such as JUnit or NUnit, or more specialized BDD tools such as RSpec (see figure 1.10).

Executable specifications like this are similar to conventional unit tests, but they're written in a way that both communicates the intent of the code and provides a worked example of how the code should be used. Writing low-level executable specifications this way is a little like writing detailed design documentation, with lots of examples, but using a tool that's easy and even fun for developers.

Unit testing the space shuttle

Good developers have known the importance of unit testing for a very long time. The IBM Federal Systems Division team was fully aware of their importance when they wrote the central avionics software for NASA's space shuttle in the late seventies. The approach they took to unit testing, where the unit tests were designed using the requirements and with examples provided by the business, has a surprisingly modern feel to it: